Study Questions: Systems

1. What is the difference between a **generative process** and a **generative model** in Active Inference, and which class in the `active_inference` library implements each?
2. Given a `true_A` matrix with shape `(3, 4)`, how many observation modalities and hidden states does the environment have?
3. If `true_A = np.eye(3)`, what does this imply about the environment's observation noise? What would `env.step(action)` always return?
4. Write the line of code that creates a `DiscreteEnvironment` with a 2-state, 2-observation system starting in state 1.
5. Explain why the columns of `true_A` must each sum to 1.0. What probability distribution does each column represent?
6. If `true_B` has shape `(4, 4, 3)`, how many hidden states and how many actions does the environment support?
7. What happens internally when you call `env.step(action=0)`? Describe the two sampling steps in order.
8. Why does `env.history["states"]` have one more entry than `env.history["actions"]` after T steps?
9. Construct a `true_B` matrix for a 3-state environment where action 0 is identity and action 1 is a cyclic permutation ($0 \to 1 \to 2 \to 0$).
10. What does `env.get_info()` return, and when would you use it during a simulation loop?
11. If you pass a 2-D `true_B` matrix (shape `(N, N)`) to `DiscreteEnvironment`, how many actions does the environment have? What is the implicit assumption?

12. Explain the validation checks that `DiscreteEnvironment.__init__` performs on `true_A` and `true_B`. What errors are raised if they fail?
13. Why is the initial observation generated by `env.reset()` stochastic even though you specify the initial state deterministically?
14. Design a `true_A` matrix for a partially observable 3-state system where observations 0 and 1 are ambiguous between states 0 and 1, but observation 2 uniquely identifies state 2.
15. How would you simulate a "slippery floor" environment where the intended action succeeds 80% of the time and transitions to a random adjacent state 20% of the time?
16. In code, how would you visualize the trajectory of states and observations generated by a `DiscreteEnvironment`? Which visualization function would you use?
17. What is the relationship between the `true_A` matrix in the environment and the `A` matrix in the agent's `GenerativeModel`? Under what condition would free energy be minimized?
18. If you run `env.step(action)` 100 times and collect the observations, how could you empirically estimate the `true_A` matrix from the data?
19. Explain how `env.timestep` is updated during `reset()` and `step()`. What is the invariant relating `timestep` to `len(env.history["actions"])`?
20. Describe a scenario where the generative process has 5 hidden states but the agent's generative model only has 3. What would be the computational consequences for belief updating?